

BAAZ JKBOSE

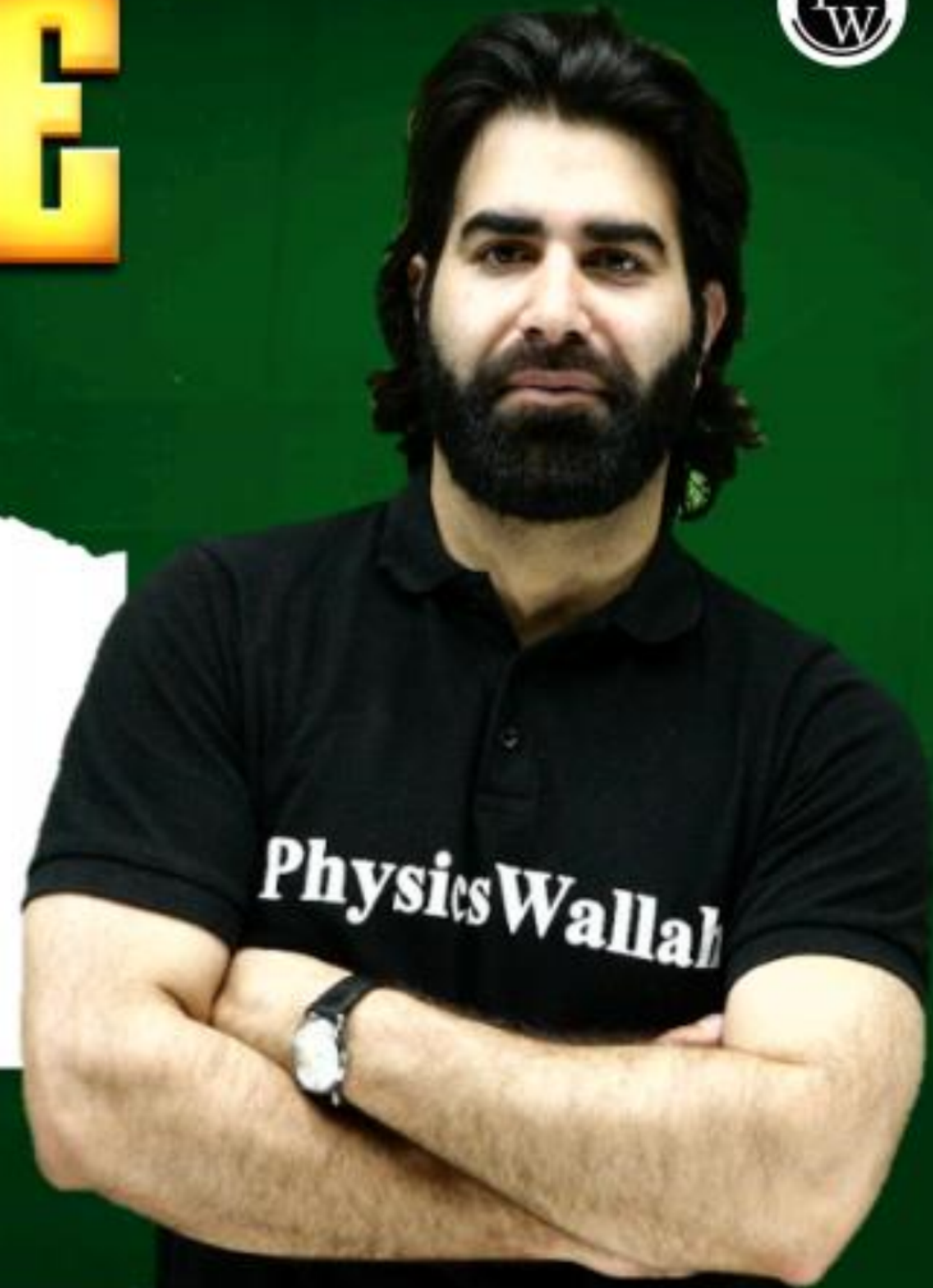
2026 CLASS-11TH

Locomotion and
Movement

LECTURE-3

Zoology

Dr Adil Altaf sir



Topics To Be Covered

1

Mechanism of muscle contraction, types of skeletal muscle

Revision

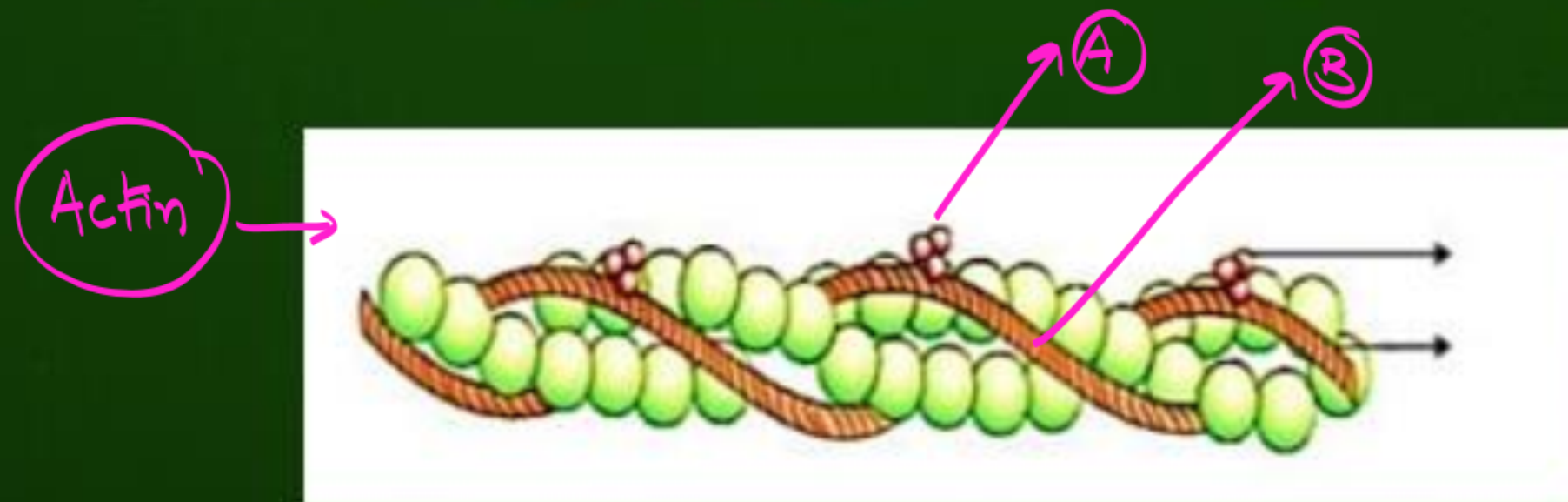
- Actin.
- Sarcomere.
- Myofibril.

✓ → A Band = Another name = ① Dark Band.

✓ → Z line = ② Centre of I Band.
(where)

✓ → M-line = ③ centre of A Band.

✓ → During Contraction, Length of Sarcomere ④ (Decrease).



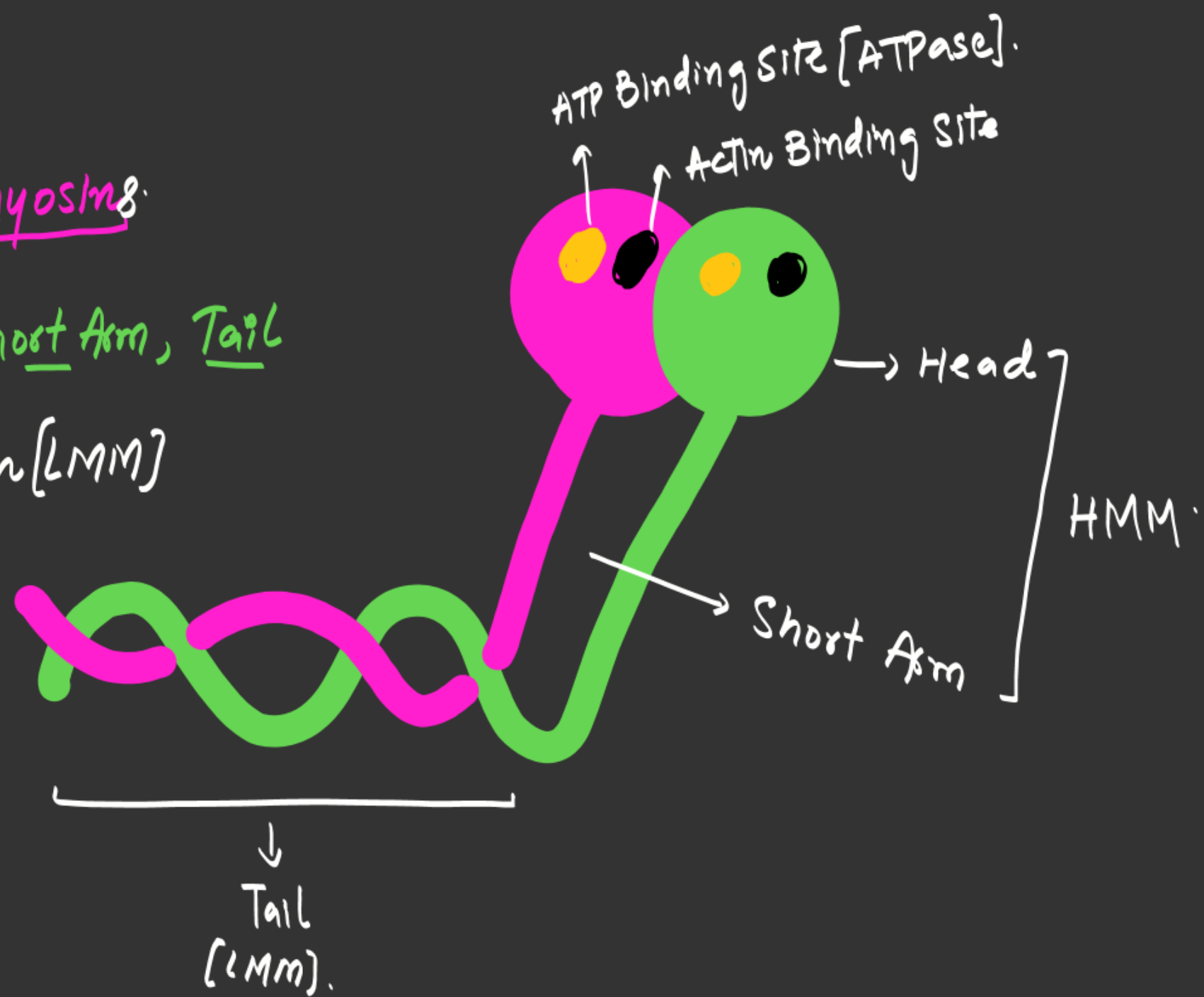
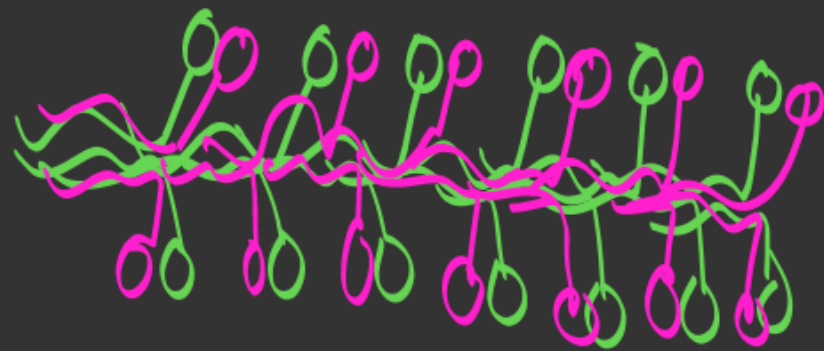
Structure of Myosin.

↳ Myosin is Made of Meromyosins.

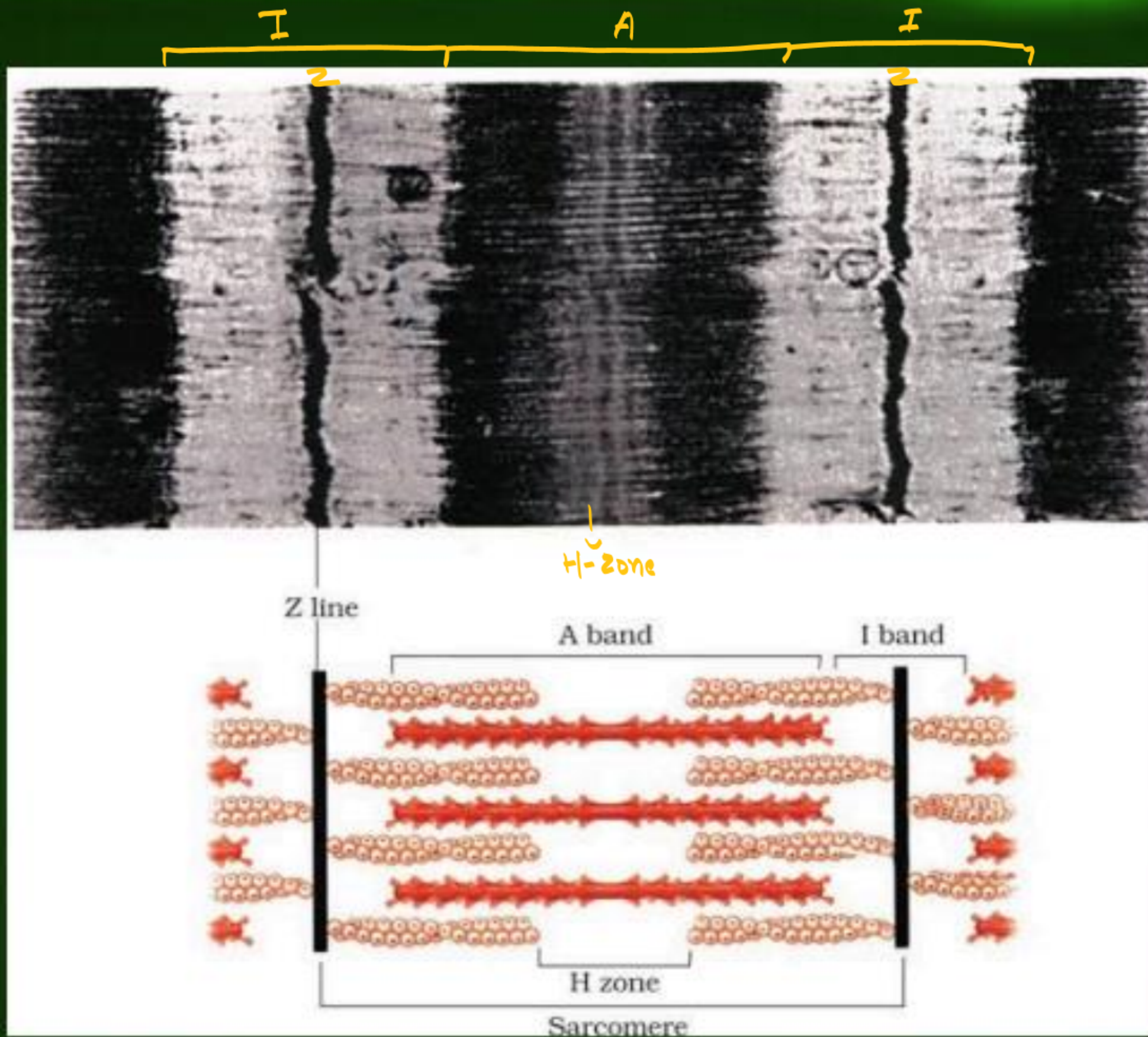
↳ Meromyosin has Head, Short Arm, Tail

↳ Tail is called = Light Meromyosin [LMM]

↳ Head & Arm = Heavy Meromyosin (HMM).

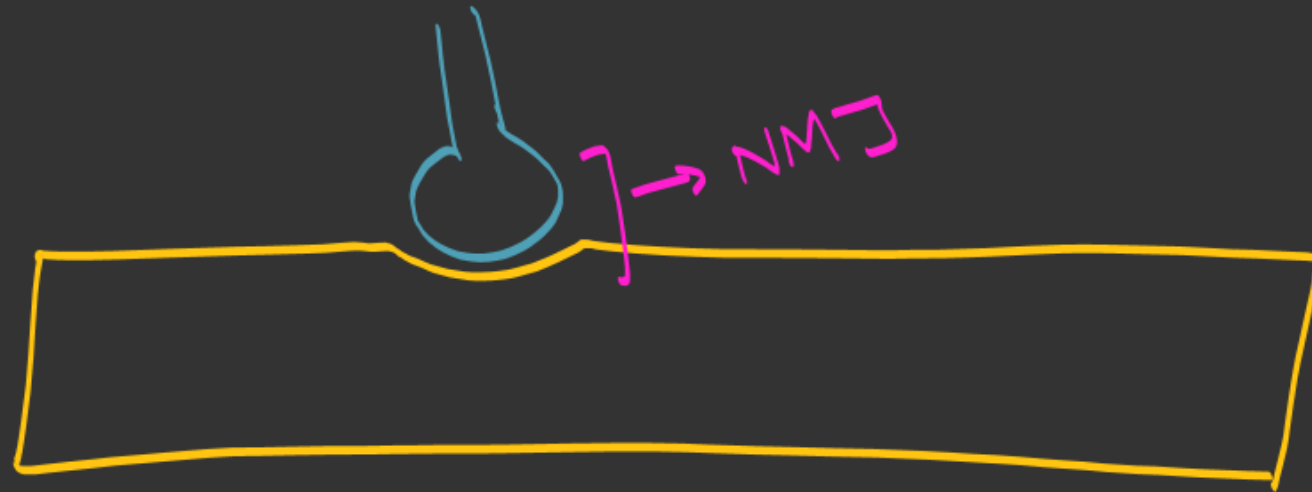


Myosin → Head & Short arm
are protruding outwards
→ Head Attaches to Actin and forms Cross Bridge

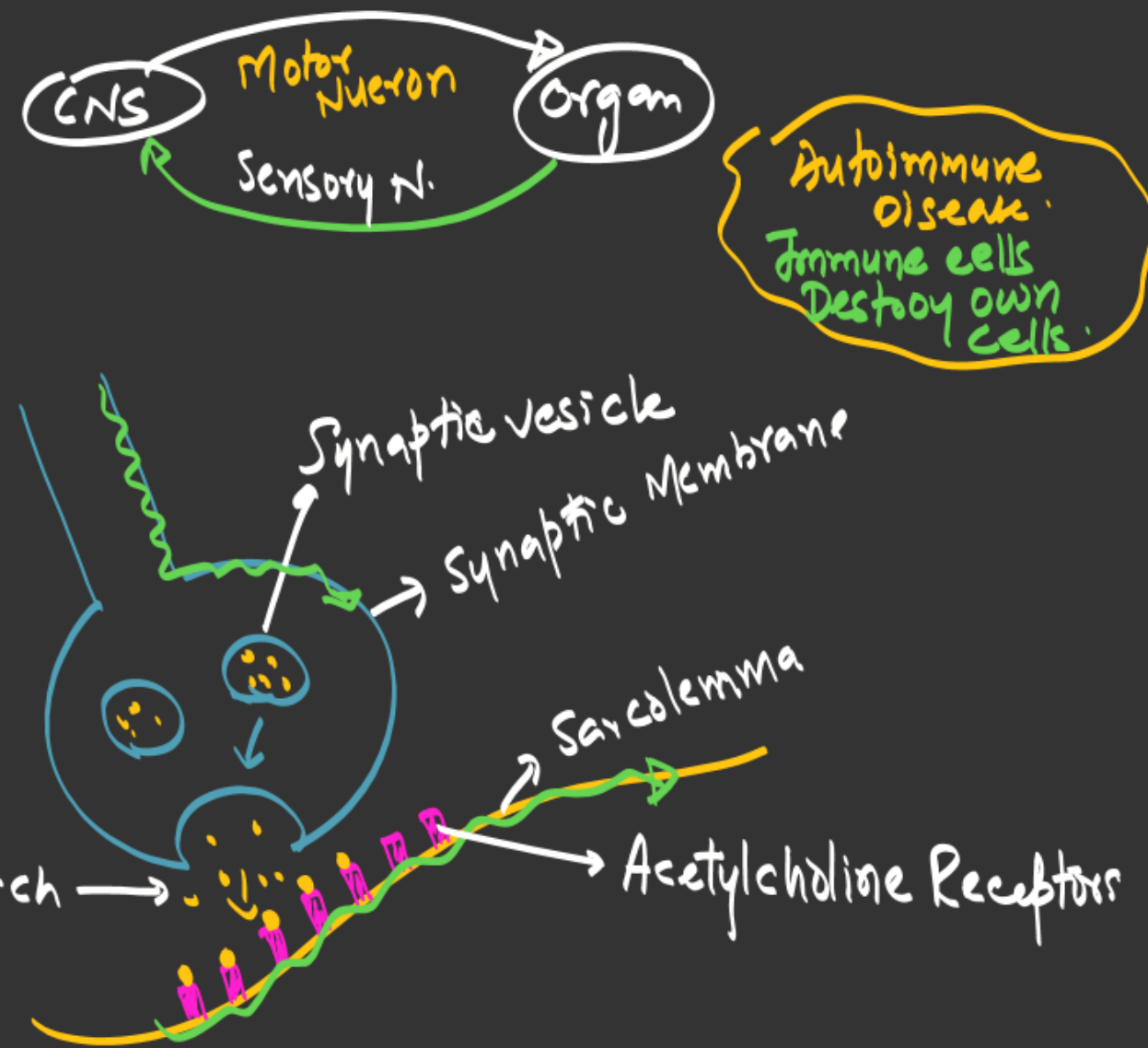


Neuro Muscular Junction.

- It is a Junction Between the Axons of Motor Neuron and Sarcolemma of Muscle fibre.



- This is Also k/as Motor End Plate.
- Signal / Impulse From Neuron is Transferred into Sarcolemma at NMJ. (Synapse).
- The neurotransmitter Involved Here is Acetylcholine



Myasthenia Gravis [Grave Weakness of Muscle]

- An autoimmune Disease of NMJ.
- Antibodies Produced Against Acetylcholine Receptors.
- Muscle contraction Does not occur properly.

Mechanism of Muscle Contraction

① Signal Carried by Motor Nerve.



② Release of Acetylcholine



③ Generation of Signal in Sarcolemma.



④ Signal transferred to Sarcoplasmic Reticulum.



⑤ Release of Ca^{++} from Sarc. Reticulum.



⑥ Ca^{++} Binds to Troponin C.



⑦ Conformational changes in Troponin



⑧ Unmasking of Myosin Binding Site on Actin



⑨ Formation of cross Bridge (Actin Myosin Bonding)



⑩ Sliding of thin filament on thick filament

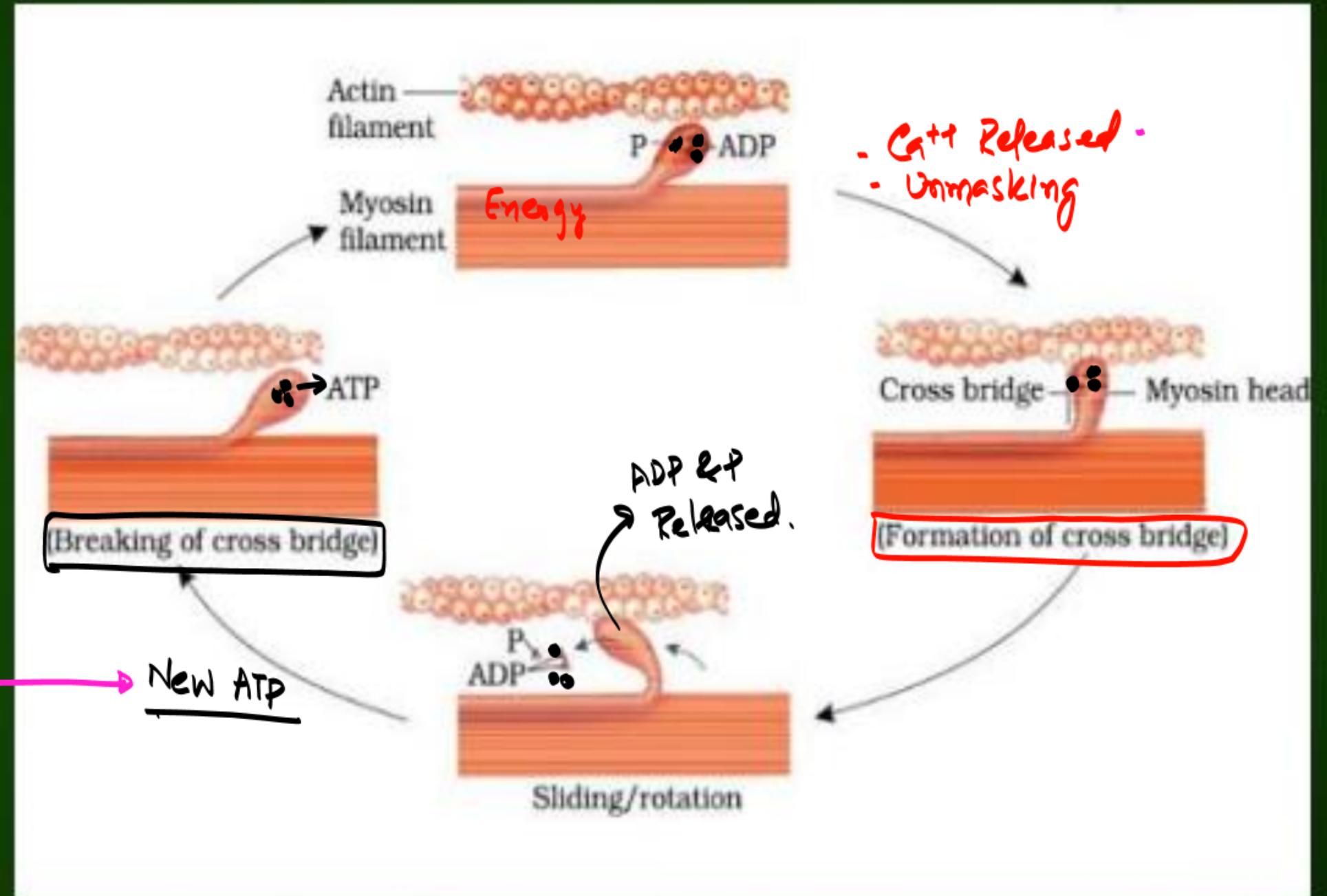
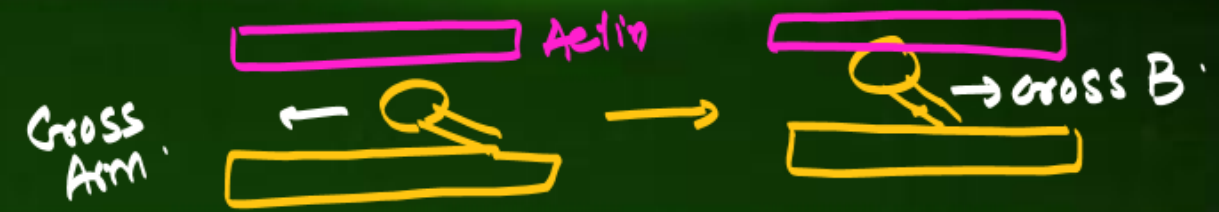


⑪ Muscle Contraction

ie length of sarcomere decreases"

Post Mortal Stiffness of Body.

Rigor Mortis



HOMEWORK



Red Muscles. vs White Muscles.

Thank
You